

Task-4: Classification Models, Evaluation Metrics & Handling Imbalanced Data

Name: Arijit Gupta

Internship: Artificial Intelligence & Machine Learning

1. Introduction

This project focuses on binary classification using the Breast Cancer dataset. The objective is to predict whether a tumor is malignant or benign using machine learning models. The implementation includes Logistic Regression and Decision Tree classifiers, evaluation using confusion matrix, precision, recall, F1-score, ROC curve, AUC score, and handling class imbalance using class weights.

2. Confusion Matrix & Metric Explanation

A confusion matrix evaluates classification performance using four components: • True Positive (TP): Model correctly predicts malignant tumor. • True Negative (TN): Model correctly predicts benign tumor. • False Positive (FP): Model predicts malignant but tumor is benign. • False Negative (FN): Model predicts benign but tumor is malignant. In medical diagnosis, False Negatives are extremely dangerous because missing a cancer case can delay treatment and risk patient life.

Accuracy alone is insufficient because in imbalanced datasets a model may achieve high accuracy by predicting the majority class only. Therefore, precision, recall and F1-score provide better insight into model performance.

3. Metric Selection for Medical Diagnosis

For medical diagnosis, Recall is the most important metric. Recall measures how many actual malignant cases were correctly identified. If recall is low, many cancer cases are missed (False Negatives), which can be life-threatening. F1-score is preferred for imbalanced data because it balances both Precision and Recall. It provides a single metric that reflects both false positives and false negatives.

4. Handling Imbalanced Data

The dataset has slight class imbalance. To address this, Logistic Regression with `class_weight='balanced'` was implemented. This technique assigns higher penalty to the minority

class during training, improving recall performance without manual resampling. ROC curve and AUC score were used to compare model discrimination ability independent of classification threshold.

5. Model Comparison & Final Decision

Logistic Regression is a stable linear model that generalizes well and is less prone to overfitting. Decision Tree is highly interpretable but can overfit easily, especially without pruning. Overfitting behavior was checked by comparing training and test accuracy. The Decision Tree showed higher training accuracy relative to test accuracy, indicating potential overfitting. Based on evaluation metrics (Recall, F1-score, and AUC) and stability considerations, Logistic Regression was selected as the final preferred model for medical diagnosis.

6. Conclusion

This project demonstrated the importance of selecting appropriate evaluation metrics for classification tasks, particularly in medical applications. While accuracy provides a general overview, recall and F1-score are more reliable indicators for healthcare problems. Logistic Regression provided stable and balanced performance, making it the most suitable model for this task.